It used to be that as we were climbing the mountain of nature the experimentalists would lead the way. We lazy theorists would lag behind. Every once in a while they would kick down an experimental stone which would bounce off our heads. Eventually we would get the idea and we would follow the path that was broken by the experimentalists. Once we joined our friends we would explain to them what the view was and how they got there. That was the old and easy way (at least for theorists) to climb the mountain. We all long for the return of those days. But now we theorists might have to take the lead. This is a much more lonely enterprise.14

String theorists have no desire for a solo trek to the upper reaches of Mount Nature; they would far prefer to share the burden and the excitement with experimental colleagues. It is merely a technological mismatch in our current situation—a historical asynchrony—that the theoretical ropes and crampons for the final push to the top have at least been partially fashioned, while the experimental ones do not yet exist. But this does not mean that string theory is fundamentally divorced from experiment. Rather, string theorists have high hopes of "kicking down a theoretical stone" from the ultra-high-energy mountaintop to experimentalists working at a lower base camp. This is a prime goal of present-day research in string theory. No stones have as yet been dislodged from the summit to be sent hurtling down, but, as we now discuss, a few tantalizing and promising pebbles certainly have.

The Road to Experiment

Without monumental technological breakthroughs, we will never be able to focus on the tiny length scales necessary to see a string directly. Physicists can probe down to a billionth of a billionth of a meter with accelerators that are roughly a few miles in size. Probing smaller distances requires higher energies and this means larger machines capable of focusing that energy on a single particle. As the Planck length is some 17 orders of magnitude smaller than what we can currently access, using today's technology we would need an accelerator the size of the galaxy to see individual strings. In fact, Shmuel Nussinov of Tel Aviv University has shown that this rough estimate based on straightforward scaling is likely to be overly optimistic; his more careful study indicates that we would require an